

PORTOFOLIO

Audrey Kyria Boren

Computer System Graduate

ABOUT ME

I am a fresh graduate in Computer Systems from Gunadarma University with a strong interest in Internet of Things (IoT), embedded systems, and hardware-software integration.

During my studies, I gained hands on experience in developing IoT based projects involving microcontrollers, sensors, actuators, and cloud platforms with a particular interest in connecting hardware and software to build practical solutions.

I am passionate about learning new technologies, developing my technical skills, and turning ideas into functional systems. While my primary interests are in IoT and embedded systems, I am also open to exploring new fields and opportunities that allow me to broaden my knowledge, gain new experiences, and grow both personally and professionally.



02

03

TECHNICAL SKILLS

- Embedded Systems & IoT
- Frontend & Web Development: HTML5, CSS3, JavaScript (ES6+), Bootstrap, Responsive Web Design, Leaflet.js (GIS/Mapping), REST API Integration.
- Mobile & Cloud Infrastructure: Flutter (Dart), Firebase Realtime Database, Vercel Deployment, PythonAnywhere, Cron-job services.
- UI/UX & Product Design: Mobile-First Layout Design, Wireframing, UI Prototyping, User Experience Design.

MY PROJECTS

04

IoT & AI-Based Water Pump Control System for Commercial Housing Optimization

Project Overview Developed an integrated IoT and AI-driven water pump control system connected to a web application to optimize water distribution and energy management in commercial housing (boarding/rented houses). The system leverages AI algorithms to regulate water usage efficiently while automatically calculating and splitting electricity costs per user in real-time (IDR currency).

During a 2-week operational trial, the system successfully optimized pump operations and reduced overall electricity consumption by 12.95%.

Key Features & Impact

- **Smart Energy & Cost Optimization:** Integrated machine learning logic to adjust pump cycles, resulting in a 12.95% reduction in electricity usage.
- **Automated Cost Splitting:** Real-time metering and proportion calculation of electricity costs per tenant converted directly into Indonesian Rupiah (IDR).
- **Real-Time Water Monitoring:** Ultrasonic-based liquid level tracking for water storage tanks (toren) accessible via a web dashboard.

- **Predictive Analytics:** Forecasting features to estimate daily and monthly electricity expenditures for water usage.

Tech Stack & Architecture

- **Hardware:** ESP32, Wemos D1 Mini, JSN-SR04M Ultrasonic Sensor (Water-resistant), PZEM-004T AC Power Meter Module, ANT-801S Vibration Sensor, SSR Relay.
- **Software & Cloud:** Firebase Realtime Database, Vercel, PythonAnywhere (Python Backend/AI Model Hosting), Cron-job.org, Arduino IDE (C++), VS Code (HTML, CSS, JavaScript).

Scan to see video demonstration



MY PROJECTS

05

Real-Time Geolocation Based Weather Forecast & BMKG Station Mapping Web App

Project Overview Developed a responsive real-time weather forecasting web application designed to streamline location retrieval. By leveraging the Browser Geolocation API alongside BMKG (Indonesian Bureau of Meteorology, Climatology, and Geophysics) open APIs, the application instantly delivers precise weather forecasts based on the user's device coordinates, bypassing the multi-step manual location selection required by standard portals. (Currently localized for Depok City).

Key Features & Technical Implementation

- **Automated Geolocation Detection:** Utilized the HTML5 Geolocation API to instantly fetch exact device coordinates upon site access, eliminating manual drop-down navigation (Province/City/District).
- **API Integration & Interactive Mapping:** Integrated official BMKG forecasting endpoints and dynamically mapped the nearest weather monitoring stations using Leaflet.js.
- **Responsive Data Presentation:** Processed complex weather payloads into an intuitive, lightweight, and mobile-friendly UI for seamless user experience.

Tech Stack

- **Frontend & Mapping:** HTML, CSS, JavaScript, Leaflet.js
- **API Services & Cloud:** BMKG Open Weather API, Browser Geolocation API, Vercel

<https://prediksi-cuaca-theta.vercel.app>

MY PROJECTS

06

IoT Smart Lighting System with Dual-Mode Control via Flutter Mobile App

Project Overview Developed a Smart Lighting System powered by IoT and integrated with a custom Flutter mobile application to offer flexible room lighting management. The system features a dual-mode operational architecture—Automatic (Sensor-driven) and Manual (App-driven)—connected via a real-time cloud database.

Key Features & Technical Implementation

- **Dual-Mode Operation:** Integrated an automatic sensoric mode using LDR (Light Dependent Resistor) to dynamically toggle lights based on ambient lux levels, alongside a manual override mode via the mobile interface.
- **Remote Mobile Control:** Designed an intuitive mobile UI using Flutter (Dart), allowing users to toggle light states (ON/OFF) remotely from anywhere with minimal latency.
- **Real-Time Data Synchronization:** Leveraged Firebase Realtime Database as a bi-directional communication bridge between the microcontrollers and the mobile application to ensure instant state updates.

Tech Stack

- **Hardware:** Wemos D1 Mini (ESP8266), LDR Sensor Module, Relay Module, AC Light Bulb.
- **Software & Infrastructure:** Flutter (Dart), Firebase Realtime Database, Arduino IDE (C++).

Scan to see video demonstration



MY PROJECTS

07

Arduino Based Automatic Rotary Waste Sorting System for Metal, Wet, and Dry Categories

Project Overview Developed an automated three-category waste classification system powered by an Arduino UNO. The system categorizes waste into Metal, Wet, and Dry categories using a multi-sensor array and automatically dispatches items into designated compartments via a motorized rotary mechanism.

Key Features & Technical Implementation

- **Automated Lid Access:** Integrated an HC-SR04 ultrasonic sensor to detect user presence and trigger hands-free main lid opening using an SG90 servo motor.
- **Multi-Sensor Material Identification:** Implemented an Inductive Proximity Sensor (LJ12A3-4-Z/BX) for metallic object detection alongside a Raindrop Moisture Module to classify organic/wet waste based on moisture levels.
- **Rotary Sorting Mechanism:** Programmed an Arduino UNO to control a 28BYJ-48 stepper motor (driven by ULN2003) to align the rotating sorter bin to the exact target compartment before release.

Tech Stack

- **Hardware:** Arduino UNO, Inductive Proximity Sensor (LJ12A3-4-Z/BX), Raindrop Moisture Sensor Module, HC-SR04 Ultrasonic Sensor, SG90 Servo Motor, 28BYJ-48 Stepper Motor with ULN2003 Driver Board.
- **Software:** Arduino IDE (Embedded C/C++).

Scan to see video demonstration



MY PROJECTS

08

Mental Health & Anti-Bullying Awareness Web Application (Frontend Web Development & UI/UX Mobile Design Project)

Project Overview Designed and developed LUGNA, a responsive, mobile mental health web application focused on frontend interactive experience and user-centered design. Built to combat social stigma and prevent self-diagnosis, the application prioritizes a calming visual design, seamless user flows, and accessible interface design to provide a safe space for individuals facing depression, anxiety, or bullying.

Key Frontend & Design Highlights

- **Responsive Layout** with Bootstrap Framework: Utilized Bootstrap to implement a mobile-first, grid-based layout structure, ensuring seamless responsiveness and consistency across various screen sizes.
- **Mobile-First UI/UX & Interactive Design:** Created a modern, empathetic user interface featuring customized layout structures, calming color schemes, and clean visual typography optimized for mobile devices.
- **Interactive Frontend Feature Implementation:** Built responsive, client-side interactive modules including a guided daily journal with a visual mood tracker, a custom medication reminder interface, and a diagnostic screening flow.

- **One-Touch Emergency Hotline UI:** Designed an accessible emergency interface ensuring users in crisis can instantly reach suicide prevention hotlines and emergency contacts without navigation friction.
- **Component-Based UI Architecture:** Structured reusable UI components (custom navigation bars, cards, interactive buttons, modal overlays) using clean semantic HTML, CSS, Bootstrap, and JavaScript.

Tech Stack & Focus Area

- **Primary Focus:** Frontend Development, Mobile Web Design, Product UI/UX
- **Technologies:** HTML, CSS, JavaScript, Bootstrap

Scan to see video demonstration



Audrey Kyria Boren

CONTACT INFO

Depok, West Java, ID

Audreykboren@gmail.com



09